

Potential & Kinetic Energy

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Potential energy is stored energy due to position or condition, while kinetic energy is energy of motion; greater height generally means more gravitational potential energy.
------------------	---

Round 1 - Retrieve It

1. What is kinetic energy?

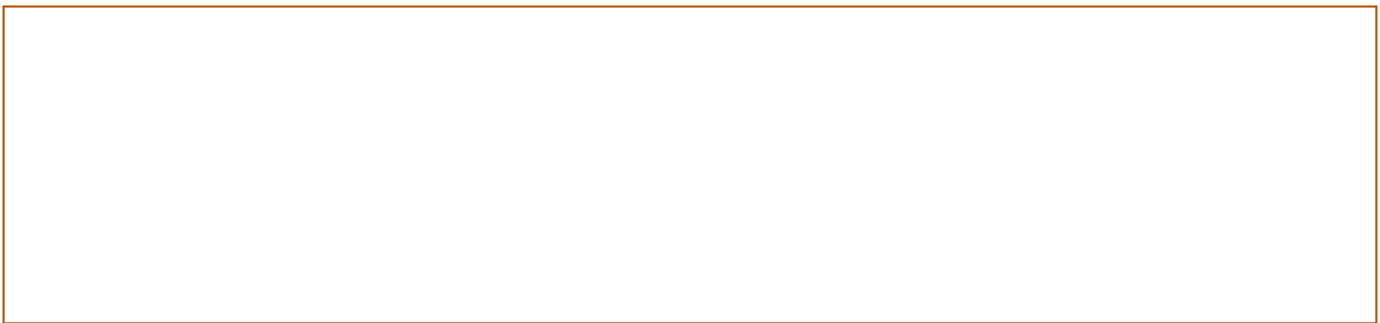
2. What is gravitational potential energy?

3. At which location on a hill is gravitational potential energy greatest, all else equal?

4. What happens to kinetic energy when an object speeds up, all else equal?

Round 2 - Sketch It

Draw a roller coaster at the top of a hill, halfway down, and near the bottom. Use PE and KE bars to show the energy trend.



Potential & Kinetic Energy - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A book sits on a high shelf. Which energy type is emphasized?

2. A rolling bowling ball is moving. Which energy type is emphasized?

3. Two identical balls move at different speeds. Which has more kinetic energy?

Round 4 - Reason from Evidence

Describe the energy transformation of a ball released from the top of a ramp. Include what happens to potential and kinetic energy.

Potential & Kinetic Energy - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Potential & Kinetic Energy in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect potential and kinetic energy to friction. Where can some mechanical energy go when friction is present?

E**Extend**

Predict how raising the starting point of a roller coaster affects its potential energy and possible speed later on the track.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Potential & Kinetic Energy

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Energy of motion.
2. Stored energy due to position/height in a gravitational field.
3. At the highest location.
4. It increases.

Sketch It - look for

Top: high PE, lower KE if slow. Downhill: PE decreases while KE increases. Bottom: lower PE and often higher KE, ignoring losses.

Apply It

1. Gravitational potential energy.
2. Kinetic energy.
3. The faster ball.

Reason from Evidence - look for

Gravitational potential energy decreases as height decreases; kinetic energy increases as speed increases. Some energy may transfer to thermal/sound due to friction.

ACE - Extend look-for

Higher starting point gives more gravitational potential energy and can allow greater kinetic energy/speed later, subject to energy losses.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.